

1. α_t 在 c_t 里

$$\begin{aligned}
 c_t &= \alpha_t^{old}(x_t, z) \frac{\beta(W(z) - 1)}{Z} (v_t - v_{old}) \\
 l(\theta) &= w_D(z) \|v_\theta - sg(v_{old} + c_t)\|_2^2 \\
 J_\theta &= \frac{\partial v_\theta(x_t, t)}{\partial \theta} \\
 \nabla_\theta l &= 2w_D(z) J_\theta [v_\theta - v_{old} - c_t] \\
 &= 2w_D(z) J_\theta [v_\theta - v_{old} - \alpha_t^{old}(x_t, z) \frac{\beta(W(z) - 1)}{Z} (v_t - v_{old})] \\
 &= -2w_D(z) \alpha_t^{old}(x_t, z) \frac{\beta(W(z) - 1)}{Z} J_\theta (v_t - v_{old}) + 2w_D(z) J_\theta (v_\theta - v_{old})
 \end{aligned}$$

2. α_t 在 MSE 外

$$\begin{aligned}
 c'_t &= \frac{\beta(W(z) - 1)}{Z} (v_t - v_{old}) \\
 l'(\theta) &= w_D(z) \alpha_t^{old}(x_t, z) \|v_\theta - sg(v_{old} + c_t)\|_2^2 \\
 \nabla_\theta l' &= 2w_D(z) \alpha_t^{old}(x_t, z) J_\theta [v_\theta - v_{old} - c'_t] \\
 &= 2w_D(z) \alpha_t^{old}(x_t, z) J_\theta [v_\theta - v_{old} - \frac{\beta(W(z) - 1)}{Z} (v_t - v_{old})] \\
 &= -2w_D(z) \alpha_t^{old}(x_t, z) \frac{\beta(W(z) - 1)}{Z} J_\theta (v_t - v_{old}) + 2w_D(z) \alpha_t^{old}(x_t, z) J_\theta (v_\theta - v_{old})
 \end{aligned}$$

- 第一项二者完全相同，是一个对 v_{old} 的 reward-induced correction
- 第二项二者相当于有两个不一样形式的 v_θ 与 v_{old} 的 anchoring 项：

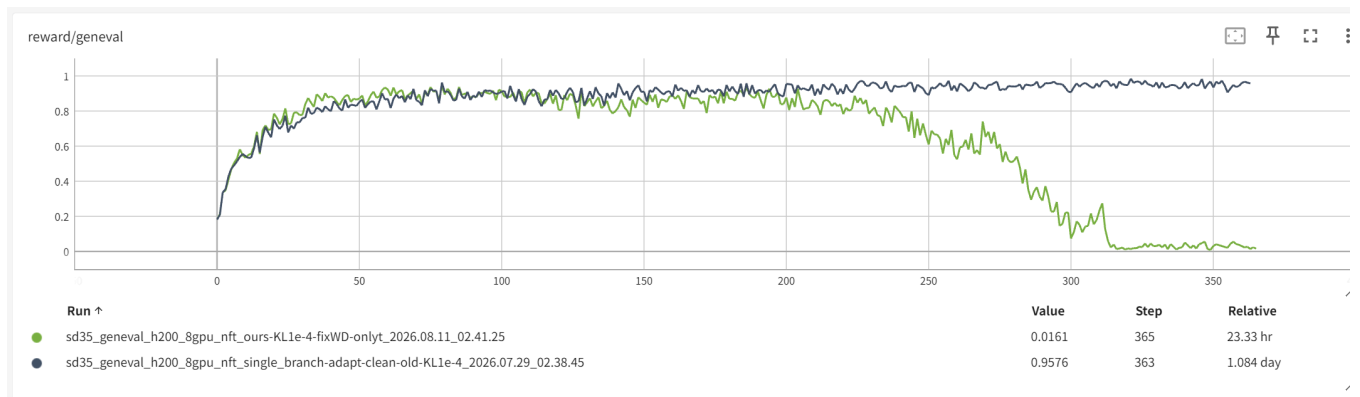
$$\begin{aligned}
 l_{anchor}(\theta) &= w_D(z) \|v_\theta - v_{old}\|_2^2 \\
 l'_{anchor}(\theta) &= w_D(z) \alpha_t^{old}(x_t, z) \|v_\theta - v_{old}\|_2^2
 \end{aligned}$$

v_θ 与 v_{old} 之间的约束关系不应该和 α_t 相关？ 所以应该采用 α_t 写在 c_t 里的形式，同时也更符合之前的推导结果
 除此之外，可以对之前的 NFT strong baseline 也测试一版 α_t 写在 $v_{target} = v_{old} + \frac{2r-1}{\beta} (v_t - v_{old})$ 的版本，即

$$\begin{aligned}
 v_{target} &= v_{old} + \frac{2r-1}{\beta} \frac{(v_t - v_{old})}{sg[mean|v_t - v_{old}|]} \\
 l(\theta) &= \|v_\theta - v_{target}\|_2^2
 \end{aligned}$$

预期不会对结果产生负面影响？

实验结果



- 绿线：

$$c_t = \alpha_t^{old}(x_t, z) \frac{\beta(W(z) - 1)}{Z} (v_t - v_{old})$$

$$l(\theta) = t ||v_\theta - sg(v_{old} + c_t)||_2^2$$

$$\alpha_t^{old}(x_t, z) = \frac{1}{sg[mean|v_t - v_{old}|]}$$

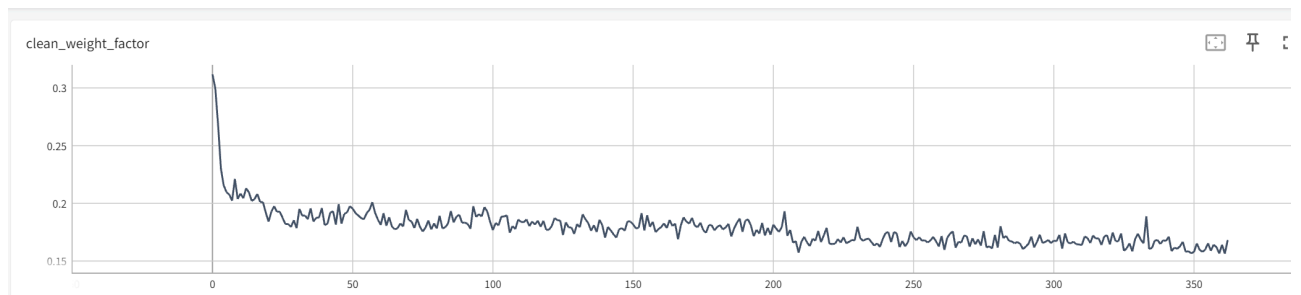
- $\beta = 1, Z = 1$

- 黑线：

$$c_t = \frac{W(z) - 1}{\beta} (v_t - v_{old})$$

$$l(\theta) = t \frac{||v_\theta - sg(v_{old} + c_t)||_2^2}{sg[mean|v_t - v_{old}|]}$$

- $sg[mean|v_t - v_{old}|]$ 值：



因此 α_t 放在 c_t 里面,相比于放在 MSE 的外面,差距在于训练中后期 $||v_\theta - v_{old}||_2^2$ 项的前面会少一个5-10之间的系数,约束强度明显更小,导致 v_θ 的优化不稳定。

